

Jacob Berger

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EDUCATION

Oregon State University Honor's College

Corvallis, OR

Bachelor of Science in Applied Computer Science

September 2022 – June 2026

Bachelor of Science in Computational Mathematics

- **Concentration:** Artificial Intelligence | **GPA:** 3.9+ | Accelerated Master's Program
- **Relevant Coursework:** Machine Learning & Data Mining, Intro to Artificial Intelligence, Analysis of Algorithms, Mathematics of Data Science (Graduate), Numerical Analysis, Linear Algebra II

RESEARCH

Honors Thesis: Dynamic Constraints for LLMs

Corvallis, OR

Researcher (Advisor: Dr. Lizhong Chen, STAR Lab)

June 2025 – Present

- Leading an honors thesis project to develop a dynamic constraint system for large language models, improving text generation for applications in computational semantics.

Senior Capstone Project: AI for Home Automation

Corvallis, OR

Project Team Member

September 2025 – Present

- Collaborating in a senior design team to integrate Large Language Models with the Home Assistant IoT platform, enabling users to control devices and create automations using natural language.

EXPERIENCE

NLP SaaS Startup

Portland, OR

Language Model and Software Engineer

March 2023 – Present

- Developed a full data pipeline for training and deploying specialized large language models.
- Led and co-authored the team's approved Phase I Small Business Innovation Research (SBIR) proposal.
- Engineered and deployed cloud infrastructure on AWS, GCP, and Vercel to train and host services.

MU Lanes and Games

Corvallis, OR

Head Mechanic

December 2022 – Present

- Managed maintenance projects and diagnostics for complex electromechanical systems.

MU Student Activities and Advisory Board

Corvallis, OR

Vice-Chairman

June 2024 – Present

- Oversaw budget preparation and facilitated strategic board meetings for OSU's Memorial Union.

SKILLS

- **Languages:** Python, C++, Java, Javascript, Matlab, x86 Assembly
- **Machine Learning & AI:** PyTorch, Open-source LLMs, LLM Fine-tuning, Controlled Text Generation
- **Software & Tools:** Linux, Git, Amazon Web Services (AWS), Google Cloud Platform (GCP)
- **Languages (Human):** Fluent in Spanish (9 years of language immersion education)

AWARDS

- **Putnam Math Competition Award:** Top 3 Finisher at Oregon State University.
- **Finley Academic Excellence Scholarship:** \$20,000 scholarship recipient for academic merit at OSU.

EXTRACURRICULARS

- **Competitive Math:** Competitor in Oregon State Putnam and COMAP Math Modeling competitions.
- **Competitive Sports:** Founded OSU's national collegiate bowling team (All-Star qualifier); served as Rose City Rowing Club team leader (2-time national qualifier).